

Principles of Robot Autonomy I

Introduction to ROS

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Attendance



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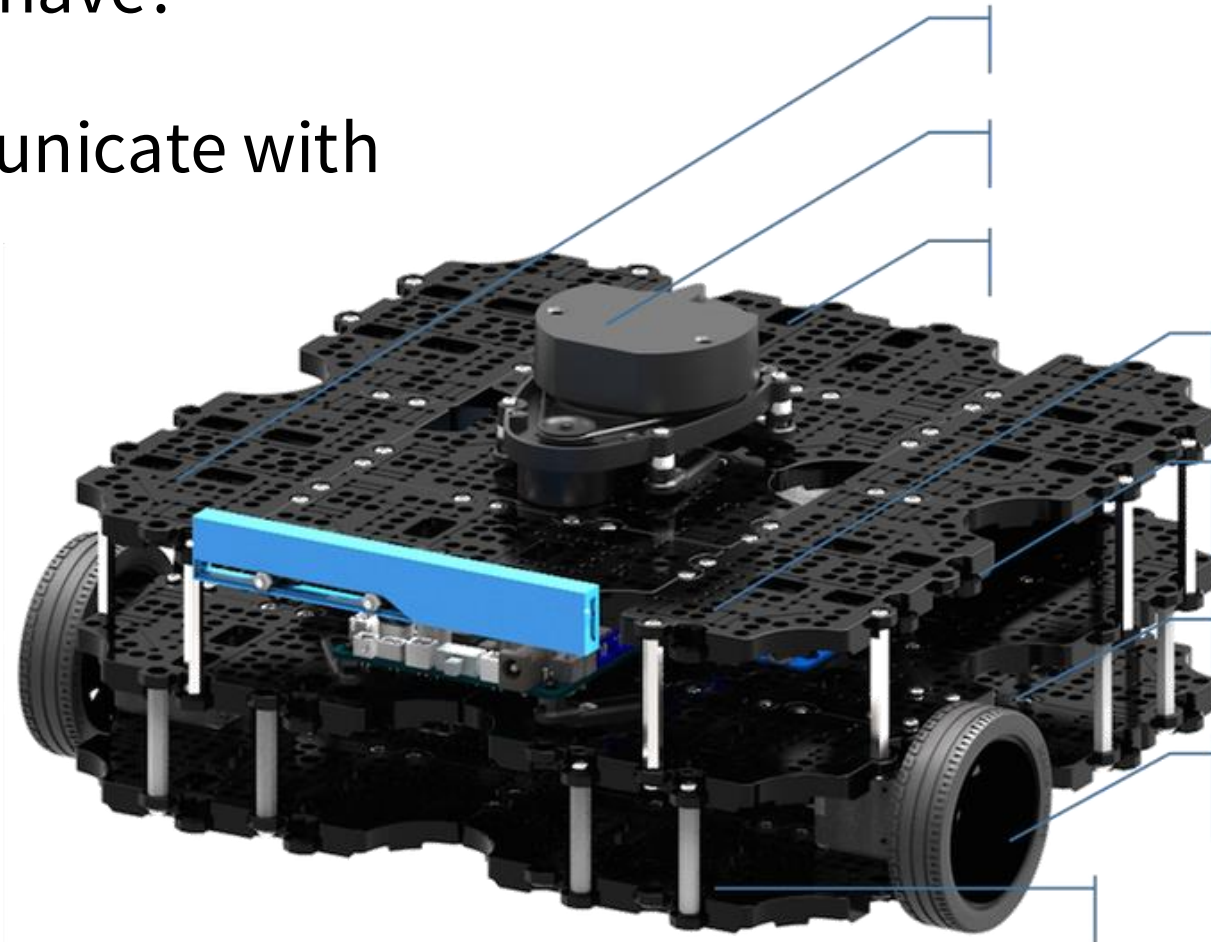
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Brain Teaser

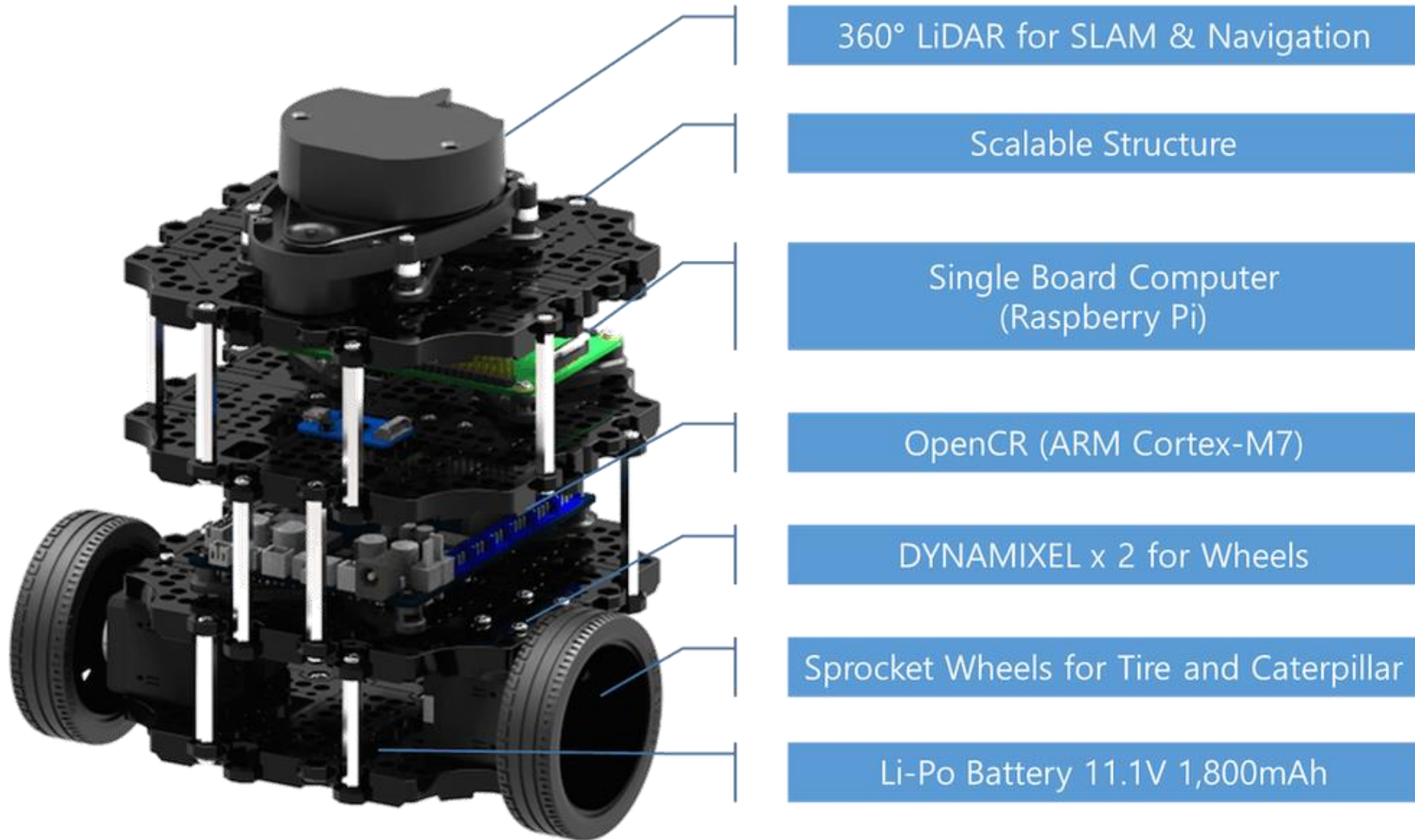
- What components does this robot have?
- Which components need to communicate with each other?
- Remember: See, Think, Act



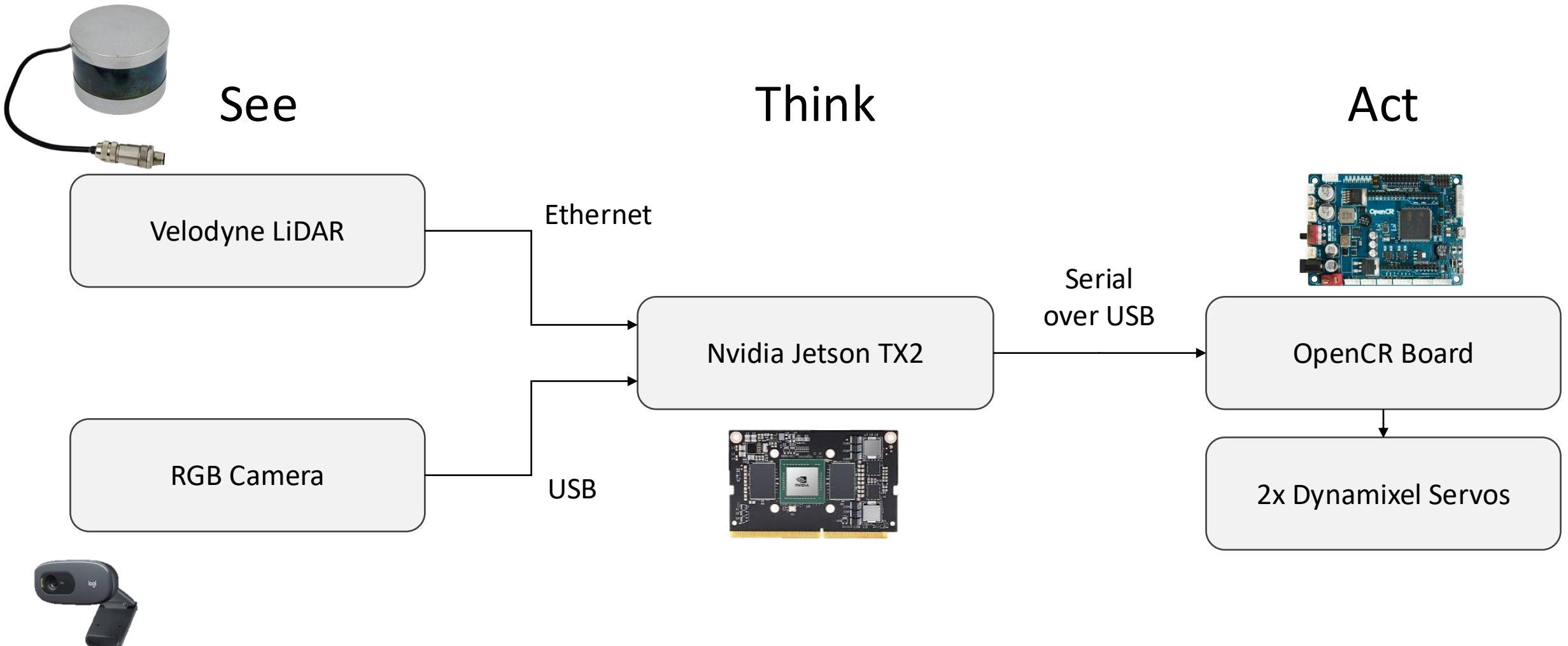
Learning goals

- Understand why and how ROS is used
- Understand the structure of a ROS workspace
- Understand how to solve problems in ROS
 - Listing available packages, nodes, topics, and parameters
 - Manually starting nodes
 - Getting information about topics
 - Send and print messages to and from topics
- Understand communication between ROS nodes
- Internalize tab completion

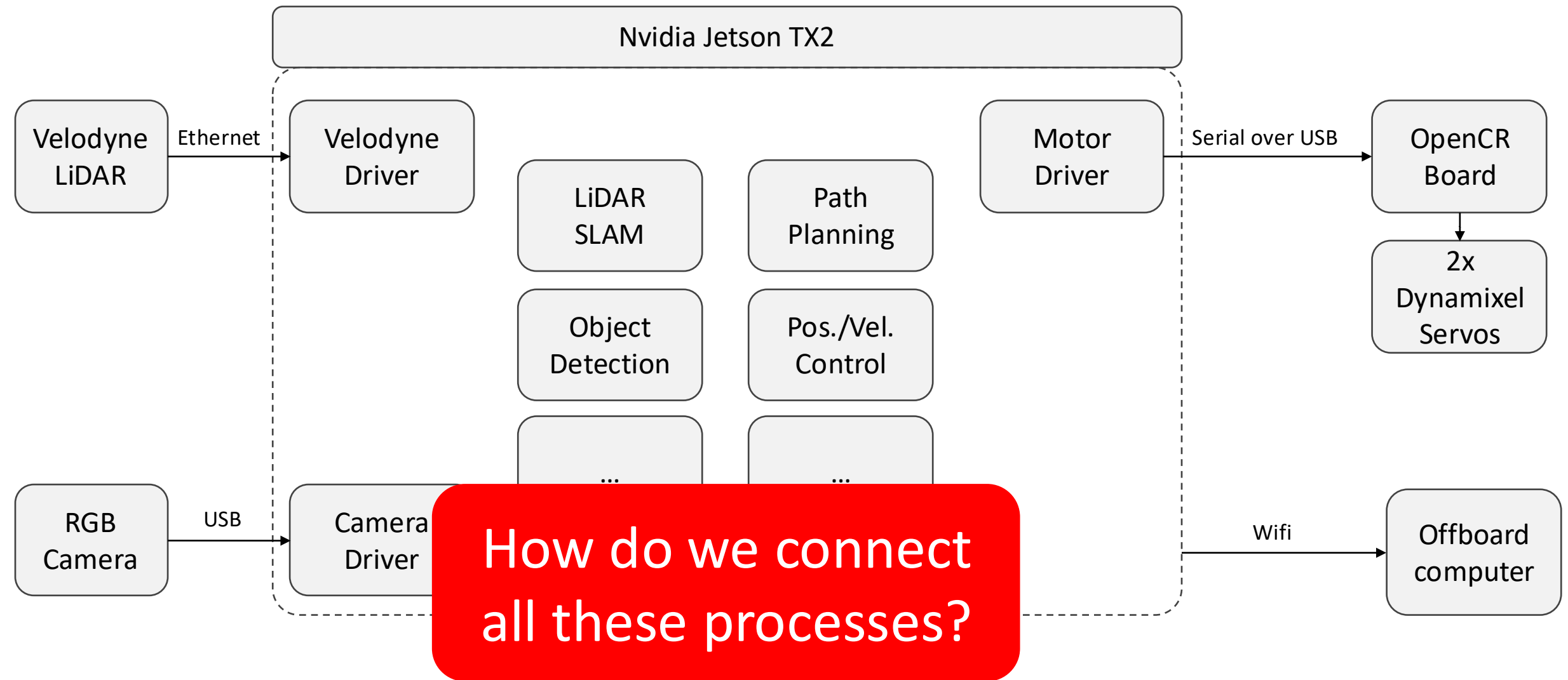
Why ROS is used



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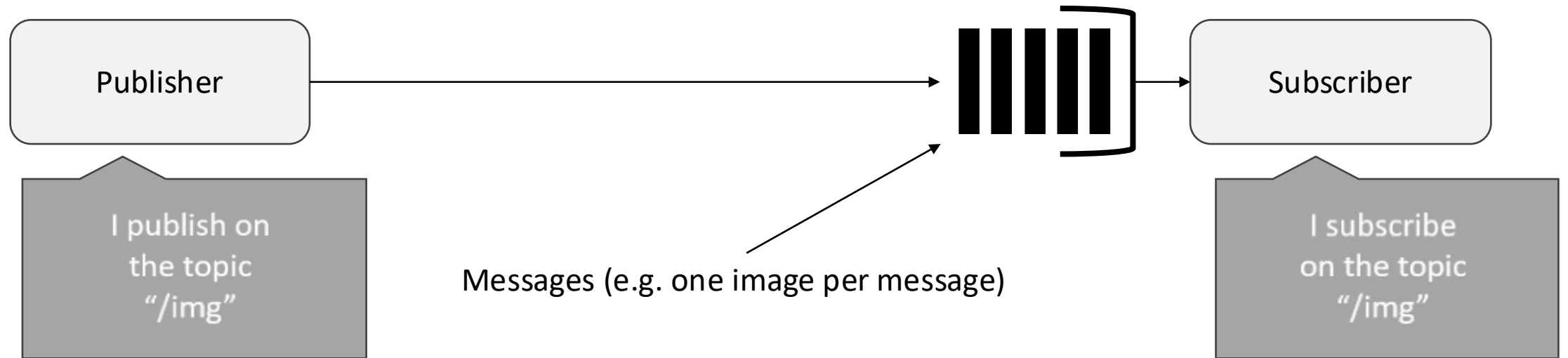


How ROS works

- ROS is a middleware that allows *nodes* (separate processes) to communicate over the computer network (IP layer)
- Communication happens through *messages* that *nodes* send and receive over *topics* (communication channels)
- Today, you will learn about *publishers* (senders) and *subscribers* (receivers)

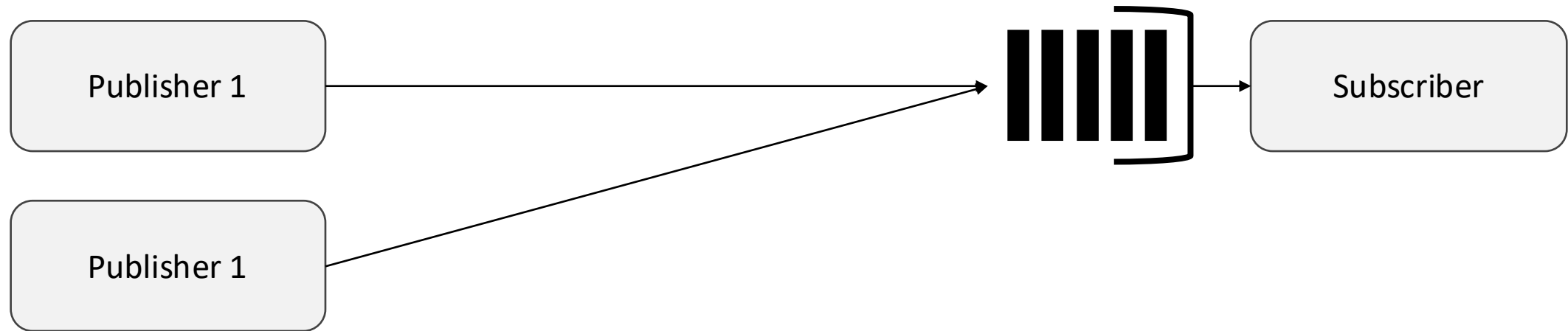
Publisher/Subscriber patterns

A publisher can publish messages to a subscriber



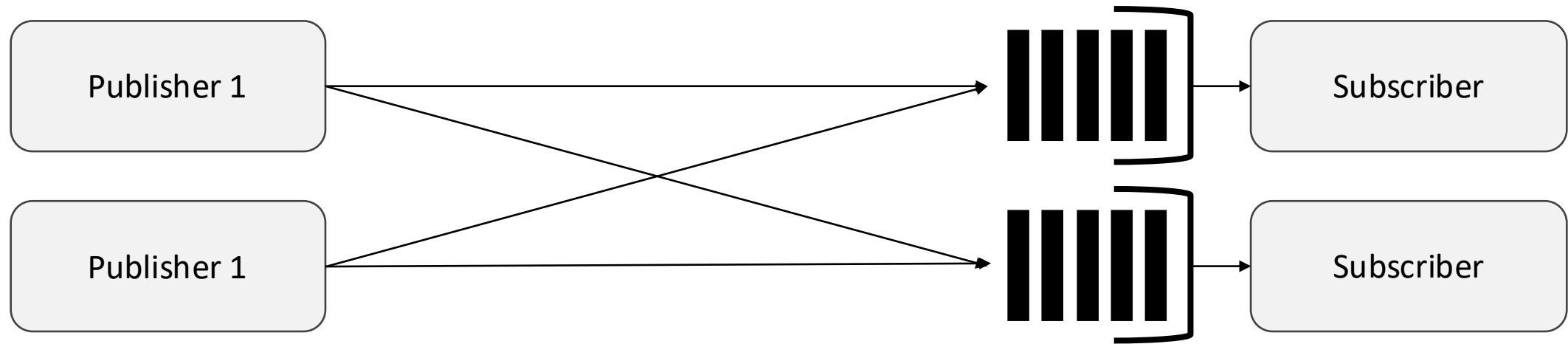
Publisher/Subscriber patterns

Many publishers can publish messages to a subscriber



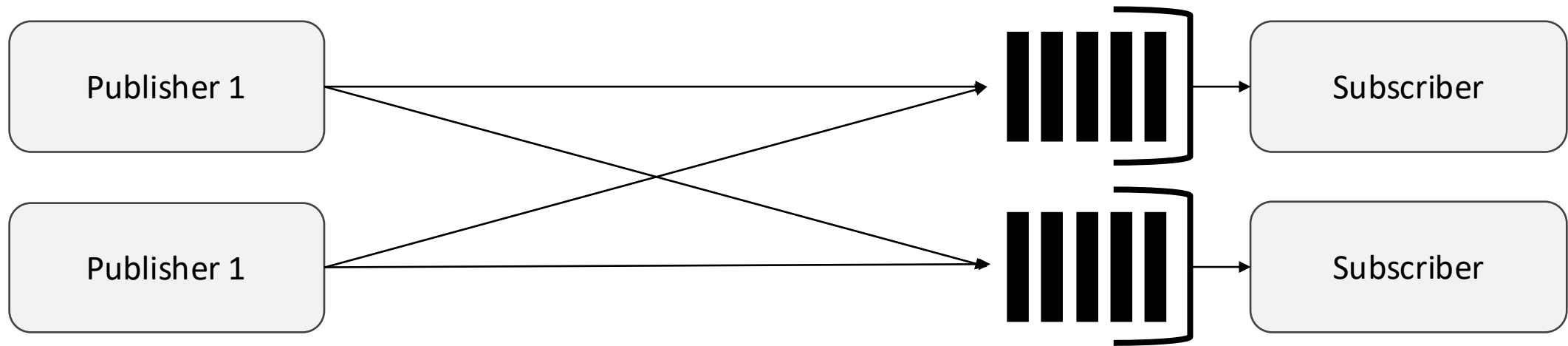
Publisher/Subscriber patterns

Many publishers can publish messages to many subscribers



Publisher/Subscriber patterns

This can be very powerful, but also creates systems that can be complex to debug!



Let's look at a simple example!

- ROS workspaces/packages
- ROS nodes – publishing and subscribing
- Launch files
- Parameters
- Logging
- Extra: Using ROS and Python inheritance

Follow along

- IF you have a laptop running a Linux distro with you:
 - <https://github.com/JakobThumm/simple-ros2-package>
- Otherwise
 - No problem!
 - You will have enough time to learn the basics of Linux and ROS in the sections on the laptops we provide.

